Autonomously Generating Potential U.S. Congressional Districts

May 7, 2026

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1. Introduction:

Gerrymandering is the process by which legislators draw unfair voting districts to maximize their chances (or the chances of their political party) of winning elections. These districts are often very irregularly shaped in order to squeeze voters into one district or another depending on many factors, such as race, income level, and prior voting history, with the intention of ensuring that a certain party has a high chance of winning an election. This disenfranchises voters by making elections less competitive; in many cases, the outcome of the election is well-known before the election actually takes place because one of the two candidates is such an overwhelming favorite.

The purpose of this project was to create a program that autonomously generates potential voting maps as a proof of concept to show that it is possible to at least begin the process of redistricting without relying completely on the biases of humans, with the end effect of reducing the prevalence of gerrymandering.

It would be naive to think that a one-semester long project that was built by one person could hope to develop a comprehensive and robust solution to this problem that is ready for real-world application. Instead, this project is a starting point for further research and development and hopes to demonstrate that there are alternatives to the current political situation in the U.S. with regards to its decennial redistricting processes. Additionally, the scope of this project has been confined to the state of Minnesota to reduce its complexity. Because each state participates in redistricting, it is assumed that the procedures and results of this project could be applied to any other state in the U.S. with minimal changes.

2. Methods:

The U.S. Census Bureau collects data at many different levels, the finest-grained of which is called a block. Blocks typically have a few dozen to a few hundred people. When working with redistricting, it would make sense to use the most fine-grained data available. However, the Census Bureau does not publish all of its data at every geographic scale. For example, income data is first reported at the block group level; block groups are collections of blocks and are therefore geographically larger. Income is an important factor to consider, so this project uses block groups as its base unit. In the state of Minnesota, there are close to 200,000 blocks and around 4,000 block groups. An additional benefit of using block groups is the decreased computational complexity of the program, which allows for more frequent testing.

The Minnesota Supreme Court has ruled that "districts shall be drawn to protect the equal opportunity of racial, ethnic, and language minorities to participate in the political process" (Order, 2021). This ruling guided this project. Because there are thousands of block groups in Minnesota, it will be necessary to first identify clusters, or hotspots, of blocks with similar characteristics, e.g. a group of blocks that all have a high population of a certain immigrant population, or a group of blocks that all have a similar income level, etc. Using these clusters, it would then be more computationally feasible to generate potential districts.

First of all, it was necessary to select which variables would be included in this process. I ultimately decided on five variables, which represent a broad variety of socioeconomic variables to describe the people living in each block group:

1. % of population over the age of 65
2. Average household size
3. Median age of the population
4. % of the population that identifies as Hispanic

5. % of the population with an annual income > $100,000

The choice to limit the analysis to these five variables was a little arbitrary but with good reason. As mentioned above, it was important to select variables that cover a large variety of the differences in population. At the same time, it was important not to select too many variables for this project because it had the potential to make the process too computationally complex. Further work should include more variables in order to build a program that is more robust and representative of the population.

There are two main sections of this project. The first calculates Local Moran's I statistics for each of the block groups based on each of the five variables listed above. Local Moran's I was chosen over global Moran's I because it allows us to determine which block groups are statistically significant high-high and low-low clusters. This is opposed to the global Moran's I statistic, which only outputs one value to describe the overall autocorrelation of the area of interest.

The second section of this project is the main section, and this is where the voting districts are generated.

The approach for building voting districts works as follows:

1. Select a starting block group. This can be arbitrary, but I selected the southwestern most point of the state.
2. Use a greedy approach to add block groups to the expanding district. This approach is rather rudimentary, and a deeper analysis of it can be found in the Limitations section of this paper.
    a. The results of the current block group's Local Moran's I tests are compared with the corresponding results of the block group's neighbors.

The goal is to find the neighboring block group that is most similar in socioeconomic characteristics to the current block group.

b. A score is calculated for each neighbor, starting at 0. It is incremented by 1 if the neighbor is in the same quadrant (high-high or low-low) for a given variable as the current block group. Therefore, the maximum score is 5, because there are 5 variables.

c. This score is calculated for each neighbor. The neighbor with the highest score is selected to join the voting district, and then the algorithm continues.

d. This process proceeds until the voting district has grown in size roughly equal to 1/8th of the total population of Minnesota. Minnesota currently has 8 voting districts, so each one needs to contain about 1/8th the population of the state.

e. Then, a new block group is chosen - one that is not already a part of a voting district - and a new district is generated.

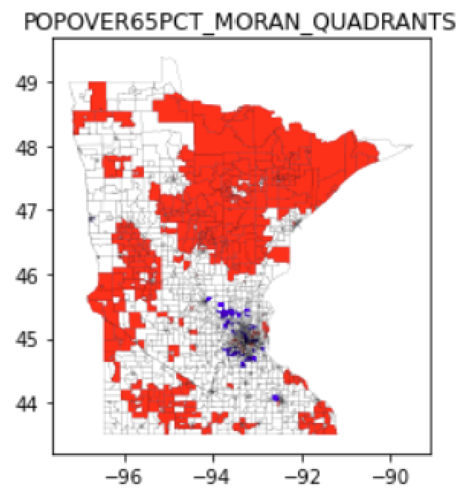
f. This process is repeated 8 times to generate 8 voting districts.

3. Data and Analysis:

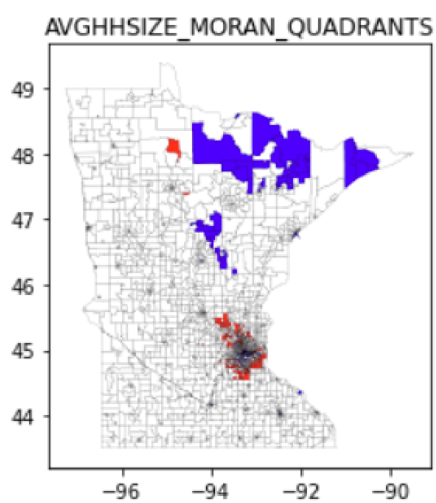
Data was collected from two locations. The first dataset contained geographic boundaries of all the block groups in Minnesota, along with dozens of socioeconomic variables (*U.S. Census*). The second dataset contained income data for each block group (*Explore census data*). These two datasets were joined together to facilitate the rest of the project.

Local Moran's I tests were run on the entire state of Minnesota for the five variables listed above. Then, the statistically significant (p = 0.05) block groups categorized as high-high (red) and low-low (blue) were plotted to visualize the results. The x- and y-axis display longitude and latitude, respectively.

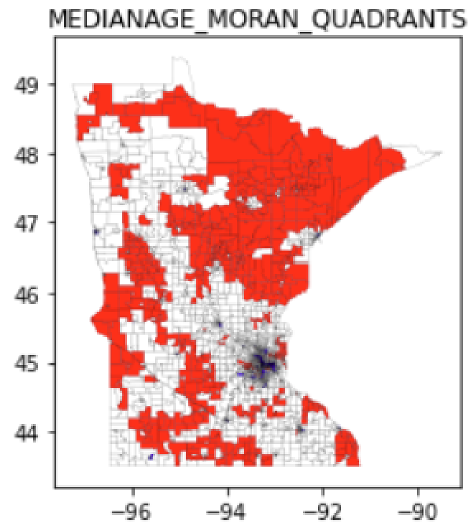
1. % of population over the age of 65



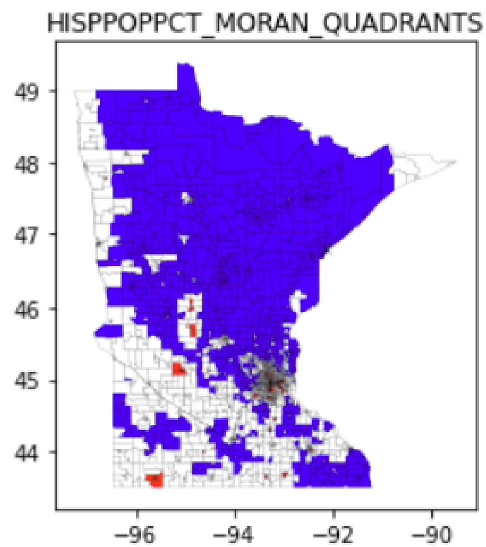
2. Average household size



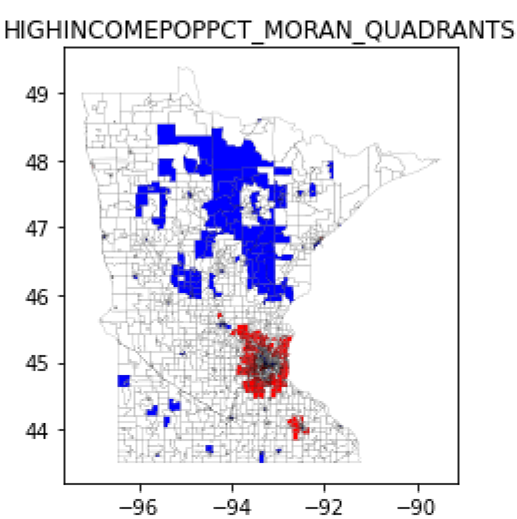
3. Median age of the population



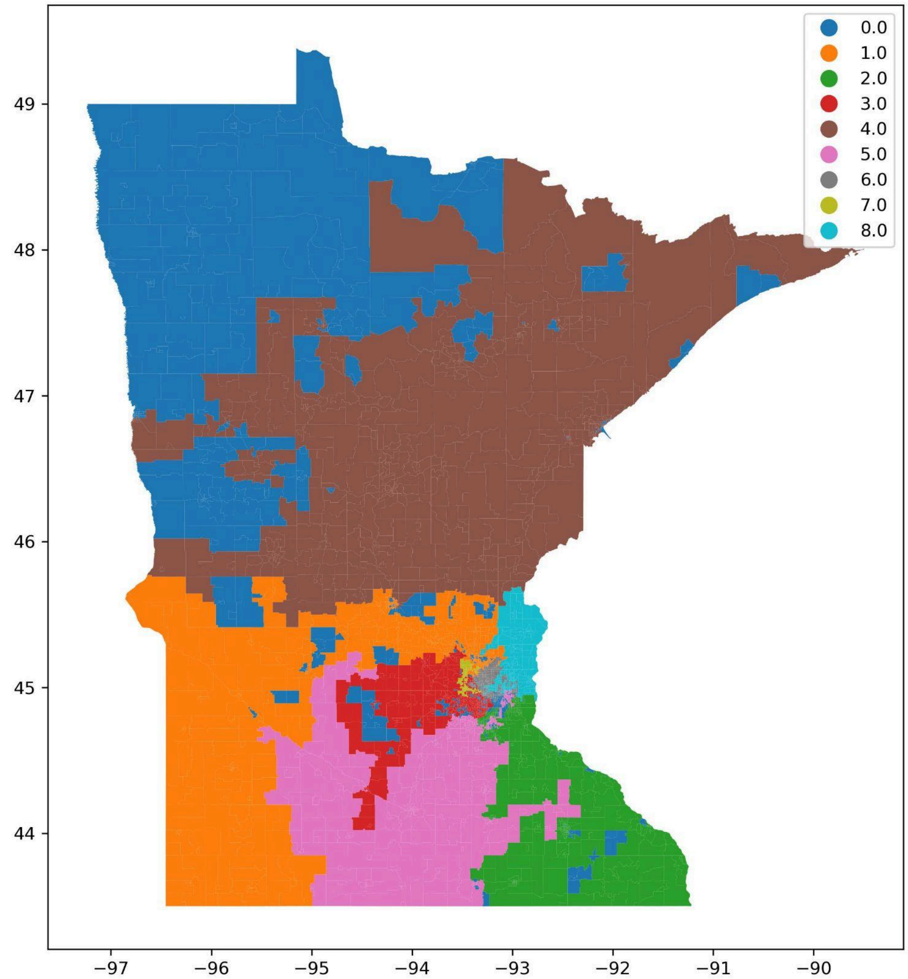
4. % of the population that identifies as Hispanic



5. % of the population with an annual income > $100,000



The district-generating algorithm outputted the following map:



It is plain to see that there were many block groups that were not selected to be a part of any voting district. These block groups are labeled as voting district 0 (dark blue) in the above plot. A discussion of this issue follows in the Limitations section.

4. Limitations of this Approach:

There are two major drawbacks of this project and this approach to generating voting districts:

1. Drawbacks of the greedy algorithm

   The greedy algorithm, which only ever looks one step ahead and selects the neighboring block group that is most immediately promising, suffers from the fact that it has very little information to work with. A more robust algorithm would look multiple steps ahead - not just at its immediate neighbors but also at the neighbors of its neighbors, and so on. This broader range of vision would allow the algorithm to make smarter choices about which block groups to include in each district.

   For example, perhaps there is a large cluster of block groups that have very similar socioeconomic characteristics to the current block group, but this cluster is not a direct neighbor. The algorithm might not ever find this cluster if there is a direct neighbor in the opposite direction that is more immediately similar in characteristics.

   The second drawback to this algorithm is that it only ever considers the characteristics of the current block group. A more robust algorithm would maintain a global score of the entire voting district thus far and choose the next block group so that it most closely matches the entire district, not just its neighboring block group. This approach is, intuitively, more difficult and complex to implement, which is why the simpler approach was used given the relatively short timeframe to work on this project.

2. Problems dealing with spatial optimization

   As shown in the plot on page 7, the algorithm terminated with a significant number of block groups unclassified. This is related to the compactness problem of spatial optimization. Sometimes, the algorithm would build a voting district that included a gap, sort of like a donut with a full ring surrounding a few block groups in the center that were not a part of the district. Because each district needs to be contiguous, these few block groups in the middle then were unable to join any other district. A more robust algorithm would identify these holes and fill them by assigning the affected block groups to the surrounding district. However, this is also computationally complex, and there was not enough time to implement this.

The second problem relates to choosing which block group to start building a district with. This algorithm ran itself into a corner by building districts in the southern half of Minnesota fairly well, but then it had no room to expand, so it terminated early for some of the districts. For example, district 7 is very small in the northwest part of the Twin Cities metro area. The algorithm ran out of contiguous block groups that hadn't already been assigned a district, which is why district 7 is so small. As a consequence of this, many block groups remained unclassified, and this can be seen in the northwestern part of the state.

It is challenging to design a clustering algorithm that satisfies both conditions of being fully enclosed (thus addressing issue 1) and appropriately spaced out from neighboring clusters such that each region is roughly the same size (population), thereby addressing issue 2. These approaches in spatial optimization are beyond the scope of this course and beyond my current abilities. Further work on this project would include significant time and attention to addressing these problems.

5. Conclusion:

Gerrymandering is a problem that continues to get worse in the U.S., as evidenced by legislative activity in many states just in the past few months. As such, it is prudent to build systems that can generate new voting districts in neutral ways. A truly comprehensive program that takes into account dozens of variables, generates thousands of potential maps, and rates them is something that would require a large research team and significant effort.

This project was designed as a small, one-person introduction to this problem and demonstrated a basic approach to solving it. It is acknowledged that much of the approach is rudimentary and would need significant improvements in order to be used in any real-world scenarios. However, the project also demonstrated that approaching the problem of gerrymandering from a geostatistical and computational perspective is possible, and it shows early but optimistic promise in its results. This project was built by one person over the course of roughly three months while working on many other projects in different areas. It is not hard to imagine the possibilities were a fulltime group of experts to take on this work and advance it outside of the classroom.

The code for this project can be found at the following public repository: https://github.com/olsonalec/GEOG5531/tree/main

References

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